Lesson plan · for the teacher

Keep it running and secure

IoT · Level 4 · Lesson 3

Time
40 min

Materials
The board and chalk, or large paper and markers; Sticky notes or slips of paper; A one-page guide template: what it does, how to check it, the battery, whom to call; Optional: a real sensor node, or a photo of one, to point at its parts

Learning goals

- Plan maintenance: batteries, cleaning, recalibration, and a heartbeat that shows a silent failure.
- Secure devices: new passwords, updates, a separate network, only the data needed.
- Hand the system over so it outlives the project team.

Activity: Failure hunt

1. Warm-up (5 min): Tell the story of the garden sensor that showed 40% every day of the holidays while the seedlings died. Ask: how could we have known?
2. Teach (8 min): The upkeep rota (batteries, cleaning, recalibration), the heartbeat and last-seen time, the Mirai story and the security habits, and the handover.
3. Hunt (12 min): In groups, pupils list ways the project could fail. For each: how we would notice, who fixes it, and how to prevent it.
4. Share (7 min): Each group adds its best three to a class table on the board. Circle the silent failures: the ones nobody would notice without a heartbeat.
5. Guide (8 min): Write the device's one-page guide together: what it does, how to check it's alive, how to change the battery, whom to call.

Teach defence only: never try passwords on devices you don't own. Before class, find out who looks after the school's router, so the guide names a real person.

Discussion

- Which failure would your school notice last, and why?
- Where should the school keep the project's passwords?
- Who in the year below could join the team now?

Exit ticket

- Name one silent failure, and say how a heartbeat would reveal it.

Try it at school tomorrow
Failure hunt: Pupils list everything that could stop your sensor project: a flat battery, mud, theft, a lost password, a holiday, an attacker. For each, they write how you would notice, who fixes it, and how to prevent it. Then the class writes the device's one-page guide together.

Practice questions · for you

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1 **Ms Wanjiru runs the school's garden sensors and moves to another school next month. Everything works, and only she knows the passwords. What should she do first?**

- ◯ Write a guide, give the school the passwords, train two people
- ◯ Share one simple password with everyone, so nobody gets locked out
- ◯ Nothing yet: it works now, so it will keep working

2 **A garden node runs on two AA batteries rated 2,000 mAh and averages 2 mA. Plan with 70% of the label.**

Usable capacity: _____ mAh. That lasts about _____ hours, a little over four weeks, so change the batteries every three weeks.

Word bank: 2,000 · 1,400 · 1,000 · 700

3 **The garden dashboard has shown 38% soil moisture, the same number, since Friday. Which could explain it?**

- ☐ A wire has come loose
- ☐ It hasn't rained, so the moisture can't change
- ☐ The battery is flat, and the screen shows the last reading
- ☐ The SIM's prepaid data has run out
- ☐ The sensor has settled, so readings stay steady

4 **Match each habit to the failure or attack it prevents.**

Changing default passwords	• • • •	Attackers using a known, already-fixed weakness
Installing updates	• • • •	A hacked sensor reaching the school's records
A separate network for devices	• • • •	A dead sensor going unnoticed
Collecting only the data you need	• • • •	Malware like Mirai logging in with factory passwords
A heartbeat	• • • •	A leak exposing data you never needed

5 **Put the steps of the 2016 Mirai attack in order, to see where defences break it.**

- ☐ It installed itself on every device that let it in
- ☐ It used them all at once to flood big websites with traffic
- ☐ It scanned the internet for cameras and routers
- ☐ It tried about 60 factory passwords on each

✎ **Who at your school could keep your project running if you left tomorrow? What would they need from you?**

Check your answers

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1. **Write a guide, give the school the passwords, train two people**

   Systems fail on ordinary things, like a flat battery or a forgotten password; with one person and no guide, the first one ends the project.

2. **Usable capacity: 1,400 mAh. That lasts about 700 hours, a little over four weeks, so change the batteries every three weeks.**

   70% of 2,000 is 1,400 mAh, and 1,400 ÷ 2 = 700 hours, about 29 days.

3. **The battery is flat, and the screen shows the last reading; A wire has come loose; The SIM's prepaid data has run out**

   Soil dries every day as water evaporates and plants drink, so a number that doesn't move for days is a warning sign.

4. **Changing default passwords — Malware like Mirai logging in with factory passwords**
   Installing updates — Attackers using a known, already-fixed weakness
   A separate network for devices — A hacked sensor reaching the school's records
   Collecting only the data you need — A leak exposing data you never needed
   A heartbeat — A dead sensor going unnoticed

   Each habit closes one door. Together they mean an attacker has to get past several defences, and a failure is noticed in hours, not weeks.

5. **Top to bottom: 3, 4, 1, 2**

   Changing default passwords breaks the chain at step two: Mirai never gets in.